João Eduardo Pereira dos Santos

Senior Python Engineer | Backend, Cloud & AI/LLM Applications

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Senior Python Engineer with **16+ years of software engineering experience** building resilient backend services, REST APIs, enterprise integrations, and data processing solutions. Strong background in **Python**, **SQL**, relational databases, and cloud-based applications. Currently expanding into AI Engineering through hands-on work with **LLMs**, **Retrieval-Augmented Generation (RAG)**, embeddings, and AI agents. Experienced in delivering maintainable enterprise solutions across **Critical TechWorks**, **EY**, and **UNICAMP**.

PROFESSIONAL EXPERIENCE

Python Full Stack Developer • Critical TechWorks (BMW Group) Nov 2025 – Present | Porto, Portugal

- Architect and integrate Python backend microservices, REST APIs, and automotive data integration workflows.
- Deploy and maintain containerized services using Docker and Kubernetes within AWS cloud environments.
- Implement automated CI/CD pipelines, regression test suites, and agile delivery practices (Scrum).
- Optimize complex SQL queries, data access layers, and enterprise system connectivity for high availability.
- Collaborate with international engineering teams across Portugal and Germany delivering reliable software solutions.

Python Full Stack Developer • Ernst & Young (EY) Dec 2022 – Nov 2025 | Brazil / Hybrid

- Developed a Python-based Retrieval-Augmented Generation (RAG) prototype for semantic search and Q&A over internal document sets.
- Engineered document ingestion, chunking pipelines, vector embeddings, and LLM query-response workflows.
- Delivered enterprise data integration pipelines for SAP migration projects, including extraction, transformation, and validation.
- Developed and maintained ETL workflows using IBM DataStage and custom Python data processing scripts.
- Authored complex SQL transformations, automated data validation routines, and operational reporting dashboards.

Senior Systems Engineer & Python Developer • UNICAMP Apr 2008 – Dec 2022 (14+ yrs) | Brazil

- Engineered scalable web applications, RESTful APIs, and administrative platforms using Python (Django, Flask) and SQL.
- Architected and delivered core administrative and academic portal backends serving over 40,000 university users.
- Designed, optimized, and administered mission-critical relational databases (PostgreSQL, Oracle, MySQL).
- Spearheaded digital transformation initiatives, DevOps automation, and Linux server infrastructure management.
- Automated complex institutional data extraction, BI reporting pipelines, and routine operational workflows.

Earlier Career: *Digital Games Programming Instructor* at SESC Santana (2016–2019; Python, Lua, game logic) • *Software & Data Analyst* at Mercer Brasil (2005–2008; SQL Server, ETL, database integration) • *Technical Support* at Atento (2001–2002).

TECHNICAL SKILLS

Backend Engineering

Python | FastAPI | Flask | Django | REST APIs | AsyncIO

AI & LLM Applications

LLM Integration | RAG Architectures | Doc Ingestion & Chunking | Embeddings | Semantic Retrieval | Prompt Engineering | AI Agents | Vector Databases

Databases & Data Engineering

PostgreSQL | Oracle DB | MySQL | SQL Server | SQL Optimization | ETL / ELT | IBM DataStage | SAP Migration

Cloud & DevOps

AWS | Docker | Kubernetes | Git | CI/CD | Jenkins | Linux

Architecture & Practices

Clean Architecture | Microservices | Enterprise Integration | Unit Testing | Agile / Scrum

EDUCATION & TRAINING

B.S. Computer Engineering
UNIVESP (Brazil)
2019 – Present (In progress)

Data Processing Technologist
FATEC-SP (Brazil)
2003 – 2007

Building Web Apps in Django
University of Michigan

Full Stack Developer Program
PUC-Rio

LANGUAGES

English	C2 Proficient / Mastery
Portuguese	Native